

Kai Cheng

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RESEARCH INTERESTS

Robot Perception, 3D Vision, Dexterous Manipulation, World Models, Representation Learning

EDUCATION

Zhejiang University Aug. 2024 – Present
B.Eng. in Mechanical Engineering, School of Mechanical Engineering; GPA: 3.57/4.00 Hangzhou, China
Relevant Coursework: Machine Learning, Data Structures & Algorithms, Automatic Control, Mechanisms & Machine Theory.

RESEARCH EXPERIENCE

Research Intern Mar. 2026 – Present
State Key Laboratory of CAD&CG, Zhejiang University, Supervised by Prof. Hao Chen Hangzhou, China

- Developed a stereo-video pipeline for temporally consistent 6D pose estimation in hand-object interaction scenes.
- Leveraged multi-view spatiotemporal cues from stereo video to improve pose-tracking robustness under occlusions.

Research Intern Sep. 2025 – Feb. 2026
State Key Laboratory of CAD&CG, Zhejiang University, Supervised by Prof. Zhaopeng Cui Hangzhou, China

- Worked on learning-based local planning for mobile robots, focusing on safe trajectory generation and robust navigation.
- Built training and evaluation pipelines and visualized planned trajectories in simulated navigation environments.

AWARDS & TRAINING

AI4Math Summer School: Lean4 and Mathematical Formalization Jul. 2025
Shanghai Jiao Tong University (SJTU) Shanghai, China

- Awarded “Excellent Participant” for outstanding performance in coursework and assessment.
- Gained hands-on experience with mathematical formalization using the Lean 4 theorem prover.

Mathematical Contest in Modeling (MCM/ICM) – Honorable Mention Jul. 2025
COMAP International

- Awarded Honorable Mention for a team project involving mathematical modeling, data analysis, and technical writing.

TECHNICAL SKILLS

Programming	Python, C++, MATLAB, L ^A T _E X
Machine Learning / CV	PyTorch, OpenCV, NumPy, scikit-learn
Robotics / 3D Vision	ROS, stereo vision, camera calibration, motion planning
Tools	Git, Linux, Lean 4